SYSTEM AND METHOD FOR IDENTIFYING TARGETED DATA STREAMS WITHIN AN ENCRYPTED TRAFFIC AGGREGATE USING INJECTED CRYPTOGRAPHIC ANCHORS

Reinaldo Javier Aguiar Marcano

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Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A signals intelligence system and method for identifying a specific target's encrypted data streams without decrypting the traffic. The method works by forcing the target's device to embed a unique, statistically rare string of characters or symbols, known as a "cryptographic anchor," into its outgoing plaintext data before encryption. A network monitoring system, positioned to observe large volumes of aggregated encrypted traffic, performs a lightweight scan for the known bit-pattern of the cryptographic anchor. A match allows the system to flag a specific encrypted stream as originating from the target. This method avoids computationally expensive decryption of all traffic and allows for the efficient identification and prioritization of a target's communications for further analysis or action.

Background of the Invention

The proliferation of end-to-end encryption presents a significant challenge for lawful signals intelligence and network monitoring. When all traffic on a network is encrypted, it is computationally infeasible to decrypt every data stream to find a specific target's communications. Traffic analysis can provide clues, but cannot definitively identify the originator of a specific data flow within a large aggregate of anonymized traffic.

An intelligence agency may wish to know if a person of interest is communicating about a particular topic (e.g., legal matters) or communicating at all, without needing to know the specific content of the communication. A method is needed to "tag" or "mark" a target's traffic at its source, so that it can be easily identified in its encrypted state at a downstream collection point.

Summary of the Invention

The present invention provides a system for identifying a target's encrypted data stream by compelling the target's source device to inject a "cryptographic anchor" into its data before encryption. A cryptographic anchor is a predetermined, statistically rare sequence of characters or symbols whose bit-pattern will be unique and easily identifiable even in an encrypted state.

The injection of the anchor is achieved by compromising the target's device. In one embodiment, the auto-correct dictionary of the target's operating system or application is modified. A common word is configured to be automatically replaced by a rare and non-standard equivalent, such as replacing the common two-letter article "la" with its all-caps version "LA", which is statistically unlikely to appear in normal text. The user is unaware of this substitution. When the user types a message, the anchor is automatically inserted into the plaintext.

The plaintext is then encrypted and transmitted over the network. A monitoring system intercepts the aggregated encrypted traffic from many users. It does not attempt to decrypt the traffic. Instead, it performs a simple, high-speed pattern search for the known encrypted bit-pattern of the cryptographic anchor. When a match is found, the system flags that data stream as belonging to the target. This knowledge can then be used to trigger other actions, such as logging the time of communication or dispatching surveillance assets, without ever needing to decrypt the message content.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the overall process, from anchor injection on the target device to detection at the monitoring system.

FIG. 2 is a conceptual diagram illustrating the anchor injection process via a modified auto-correct dictionary.

Detailed Description of the Invention

Referring to the flowchart in **FIG. 1**, the process begins at the Target Device (101), which has been previously compromised. A Cryptographic Anchor is injected into the plaintext data being generated by the user (102). In a preferred embodiment, this is done by modifying software on the device, such as the system's auto-correct dictionary.

The plaintext, now containing the anchor, is then encrypted by the device's standard encryption protocol (103) and transmitted over a public network. The resulting encrypted stream (104) is aggregated with the encrypted traffic of many other non-target users (105), creating a large volume of undifferentiated, encrypted data (106).

A Network Monitoring System (107) intercepts this aggregated traffic. The system does not attempt a computationally expensive decryption. Instead, a lightweight Pattern Scanner (108) performs a high-speed search for the known encrypted bit-pattern of the specific cryptographic anchor.

If the pattern is not found, the data stream is discarded (109). If the pattern is found (110), the system flags the data stream as originating from the target. This positive identification can then be used to trigger a pre-determined action (111), such as logging the target's activity or dispatching a surveillance team.

FIG. 2 illustrates the anchor injection mechanism. On a standard user device, a user types a common word, such as "la" (201). The device's auto-correct system (202), which has been modified by an outside entity, has a custom rule. This rule replaces the common word "la" with a statistically rare and non-standard bigram, "LA" (203), which becomes the cryptographic anchor. The user is unaware of this substitution. This anchor is now part of the plaintext document or message (204) and will be present in the data stream when it is subsequently encrypted. Another example is leveraging a rare symbol like the section sign (§) that is known to appear in specific document types the target works with, which can also serve as a reliable anchor.

Claims

What is claimed is:

1. A computer-implemented method for identifying a target's encrypted data stream, comprising the steps of:

    - modifying a software component on a target user's device to cause the automatic injection of a predetermined, statistically rare data anchor into a plaintext data set generated by the user;
    - encrypting the plaintext data set, including the data anchor, to create an encrypted data stream;
    - transmitting the encrypted data stream over a network;
    - at a network monitoring point, searching the encrypted data stream for a bit-pattern corresponding to the encrypted form of the predetermined data anchor; and
    - upon finding said bit-pattern, flagging the encrypted data stream as originating from the target user.

2. The method of claim 1, wherein the software component is an auto-correct dictionary.

3. The method of claim 2, wherein the data anchor is a non-standard capitalization of a common word.

4. The method of claim 1, wherein the data anchor is a rare symbol known to be associated with the target's specific activities.

5. The method of claim 1, further comprising the step of triggering a responsive action upon flagging the encrypted data stream, wherein the action does not require decryption of the data stream.

6. A system for identifying encrypted communications, comprising:

    - an injection module configured to run on a target device and to automatically insert a predetermined data anchor into plaintext data; and
    - a monitoring system, remote from the target device, comprising:
        - a network interface for intercepting encrypted data streams;
        - a pattern scanner configured to perform a search for a bit-pattern corresponding to the encrypted data anchor within the intercepted data streams; and

– a flagging module configured to identify a data stream as belonging to the target upon a positive match from the pattern scanner.

7. The system of claim 6, wherein the injection module is a modified auto-correct dictionary file.

Drawings

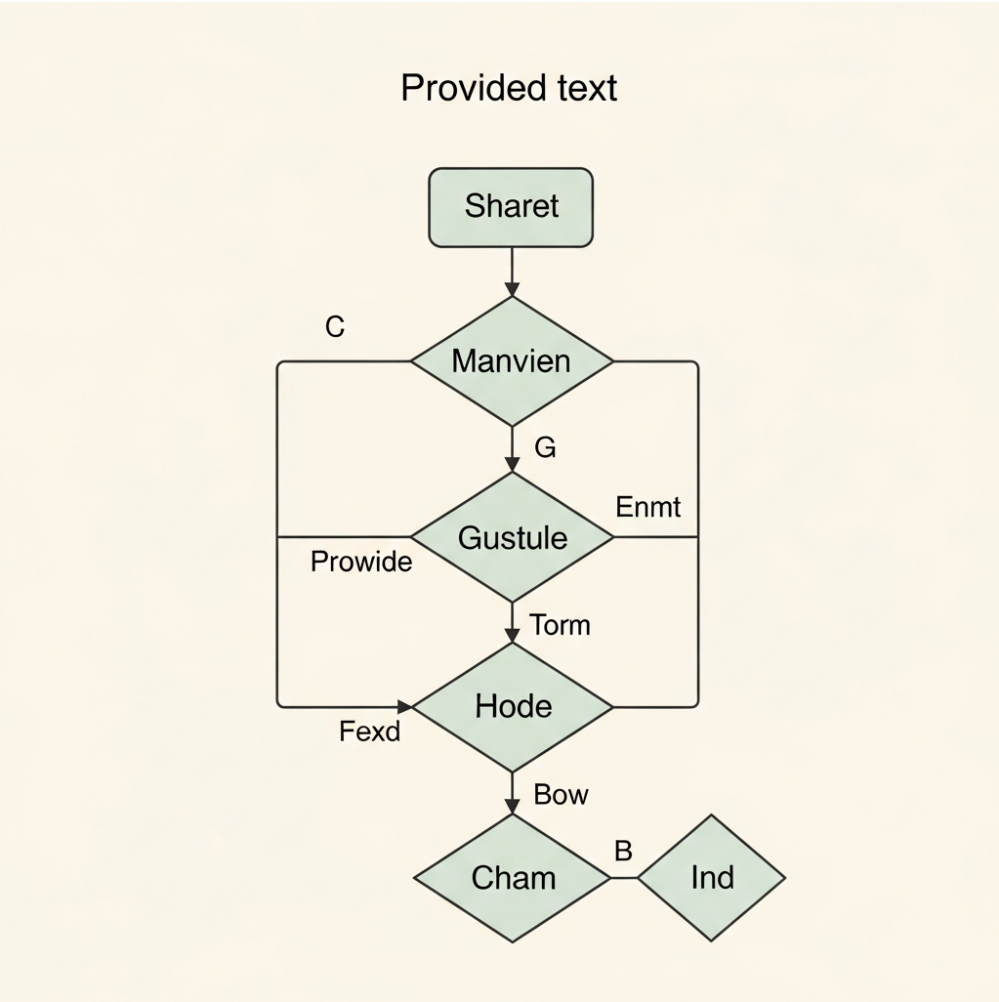


Figure 1: Flowchart of the identification process.

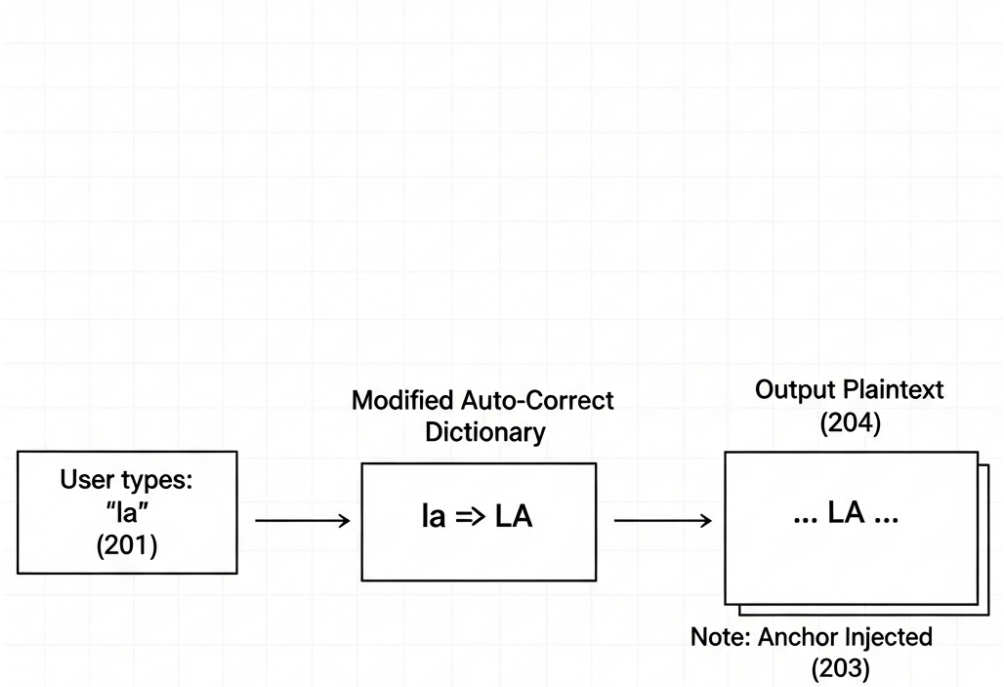


Figure 2: Anchor injection via modified auto-correct.